

Table 9. Requirements and critique traceability

Panel A. Criticism and disposition

Requirement id	Criticism	Request attribution	Disposition
A1	Correct the physical spin system	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
A2	Construct the anthracycline reaction network	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
A3	Encode a real spin Hamiltonian	Critique-author recommendation; original reviewer correspondence unavailable	partially addressed
A4	Replace CZ repetition with Hamiltonian propagation	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
A5	Correct the singlet measurement	Critique-author recommendation; original reviewer correspondence unavailable	superseded by corrected model
A6	Implement spin-selective reaction dynamics	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
A7	Add a matched classical comparison	Critique-author recommendation; original reviewer correspondence unavailable	partially addressed
A8	Add computation-time and memory benchmarks	Explicitly attributed by the critique to Reviewer 1; original correspondence unavailable	corrected with evidence
A9	Test scaling with hyperfine-coupled nuclei	Critique-author recommendation; original reviewer correspondence unavailable	withdrawn with justification
A10	Replace the T1/T2 treatment	Critique-author recommendation; original reviewer correspondence unavailable	superseded by corrected model
A11	Replace or rescale seconds-long parameters	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
A12	Rebuild the magnetic-field model	Critique-author recommendation; original reviewer correspondence unavailable	partially addressed
A13	Correct the legacy beta definition	Critique-author recommendation; original reviewer correspondence unavailable	superseded by corrected model

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Panel A. Criticism and disposition (continued)

Requirement id	Criticism	Request attribution	Disposition
A14	Resolve implementation defects in public code	Critique-author recommendation; original reviewer correspondence unavailable	superseded by corrected model
A15	Regenerate all results	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
B1	Add a chemical reaction schematic	Critique-author recommendation; original reviewer correspondence unavailable	partially addressed
B2	Add the reviewer-requested model-comparison schematic	Called reviewer-requested by the critique; original correspondence unavailable	partially addressed
B3	Add a complete circuit diagram	Critique-author recommendation; original reviewer correspondence unavailable	partially addressed
B4	Add a parameter-provenance table	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
B5	Add validation and performance figures	Critique-author recommendation; original reviewer correspondence unavailable	partially addressed
B6	Rebuild or remove the existing figures	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence
B7	Rebuild or delete the tables	Critique-author recommendation; original reviewer correspondence unavailable	corrected with evidence

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Panel B. Evidence and remaining limitation

Requirement id	Repository evidence	Remaining limitation
A1	Complete $S=1/2 \times 1$ six-state space and exact D/Q projectors.	No chemically measured encounter preparation is available.
A2	Upstream semiquinone, primary superoxide, HO_2/O_2 -minus, H_2O_2 , and loss are separate stages.	Formal semiquinone protonation remains condition-dependent; the downstream calculation is a fixed-pH pulse.
A3	Hermitian six-state Zeeman, exchange, dipolar, O_2 ZFS, and local electronic-field terms are implemented and unit documented.	Exact doxorubicin hyperfine, geometry, J, dipolar, and encounter O_2 tensors are unavailable and disabled or sensitivity-only.
A4	Classical propagation uses $\exp(L dt)$; the circuit inserts one dense Hamiltonian-derived unitary.	No product-formula circuit exists; Trotter terminology and convergence claims are withdrawn.
A5	The invalid singlet/triplet observable is replaced by exact doublet/quartet projectors and numerical expectations.	No finite-shot or measurement-gate experiment is executed.
A6	Separate kD, kQ, and escape loss channels are integrated with survival closure.	D/Q selectivity is a conditional modeling assumption; exact rates are unavailable.
A7	Independent matrix-exponential and adaptive solvers match for the same full open-system problem; coherent Qiskit matches the same dense unitary.	The coherent circuit does not implement reaction, relaxation, or escape, so no open-system circuit validation is claimed.
A8	A bounded table separately times matched coherent NumPy/Qiskit work and matched classical open-system solvers, with phase and memory scope.	Process RSS is a whole-process high-water mark; no quantum hardware, transpilation, sampling, scaling, or performance advantage is evaluated.
A9	Scaling and efficiency claims are removed; no artificial nuclear-spin model is introduced.	Exact doxorubicin hyperfine tensors are unavailable. Withdrawal is not reviewer acceptance.
A10	The old amplitude/phase partition is retired; the retained generator is explicitly phenomenological isotropic Lindblad depolarization.	It is not a measured T1, T2, or Tphi model; exact-system relaxation values are unavailable.
A11	All encounter axes use t^*k_{ref} and rates/ k_{ref} ; seconds are explicitly illustrative conversions from k_{ref} .	The reference rate is not a measured encounter clock.

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Panel B. Evidence and remaining limitation (continued)

Requirement id	Repository evidence	Remaining limitation
A12	Static Zeeman inputs in tesla are distinct from a named local electronic-field proxy in rad/s.	No stochastic magnetic-noise distribution, spectrum, correlation time, or physical field-effect prediction is implemented.
A13	The legacy beta variable and its conclusions are absent from active calculations and outputs.	No old beta boundary or trend is retained; this scope reduction is not reviewer acceptance.
A14	CZ depth, parity-as-singlet, one-kick, semiconductor calibration, and protocol-weighting code paths are retired.	The current model does not reproduce or reinterpret those obsolete outputs.
A15	Figures 1-11, Tables 1-10, captions, CSV sources, checksums, and manifests are regenerated from the corrected model.	Removed amplitude/phase, protocol, one-kick, beta, and Trotter analyses are not rerun because their claims were retired.
B1	The existing author-created Figure 1, its caption, README equations, and Table 1 document species, charges, unpaired-electron counts, multiplicities, and singlet-oxygen exclusion.	No new detailed chemical drawing was added under the restricted-imagery scope.
B2	Figure 1 plus Table 1 and Table 7 separate the full classical open-system model from the coherent-only circuit.	No new side-by-side schematic was added; the visual request was reduced to text/table evidence and is not accepted by assumption.
B3	Table 7 and the Figure 11 caption document basis indices, supplied statevector, dense unitary, duration, and numerical D/Q post-processing.	No preparation, measurement, noise, reaction, delay, or Trotter gates are executed; no new diagram was added.
B4	Complete machine-readable parameter provenance is retained with curated readable panels.	Many exact encounter parameters remain unavailable; no surrogate is activated as evidence.
B5	Numerical solver validation and coherent circuit consistency remain Figures 8 and 11; bounded runtime/memory is Table 10.	Trotter, nuclear-spin scaling, and quantum-performance figures are withdrawn with their claims.
B6	Figures 1-11 are corrected-model outputs with one CSV and caption each; Figure 1 alone is an author-created conceptual map.	All quantitative encounter coordinates remain illustrative and non-predictive.

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Panel B. Evidence and remaining limitation (continued)

Requirement id	Repository evidence	Remaining limitation
B7	Tables 1-10 are rebuilt as complete CSV sources plus readable PDF and numbered high-resolution PNG pages.	Presentation panels round values and select columns, while the linked CSV retains full precision and provenance.

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Table 9. Requirements and critique traceability

Panel C. Implementation and verification

Requirement id	Exact file	Exact code object	Figure or table	Test	Evidence status
A1	spin_chemistry.py	P_DOUBLET; P_QUARTET; initial_density	Table 1	SpinAlgebraTests; InitialStateAndHamiltonianTests	corrected with evidence
A2	spin_chemistry.py; README.md	primary_superoxide_format ion; downstream_ros_speci es_resolved	Figures 1 and 9; Table 1	StagedChemistryTests	corrected with evidence
A3	spin_chemistry.py; config s/doxorubicin_parameters. json	hamiltonian; EncounterParameters	Tables 1-3	InitialStateAndHamiltonia nTests; AuthorityAndModeTests	partially addressed
A4	spin_chemistry.py	propagate_encounter_refer ence; execute_three_qubit _unitary_circuit	Figure 11; Table 7	CoherentValidationTests; EncounterDynamicsTests	corrected with evidence
A5	spin_chemistry.py	P_DOUBLET; P_QUARTET	Figures 2, 3, and 11; Table 7	SpinAlgebraTests; CoherentValidationTests	superseded by corrected model
A6	spin_chemistry.py	propagate_encounter_refer ence; propagate_encounter _independent	Figures 2-7; Tables 4-6	EncounterDynamicsTests; I ndependentEncounterValida tionTests	corrected with evidence
A7	spin_chemistry.py; paper_analysis.py	_solver_validation; _circuit_rows	Figures 8 and 11; Tables 6 and 7	IndependentEncounterValid ationTests; CoherentValidationTests	partially addressed
A8	paper_analysis.py	_runtime_memory_benchmark	Table 10	PaperPipelineTests	corrected with evidence
A9	README.md; metadata/run_m anifest.json	SCIENTIFIC_LIMITATIONS	Table 9	DocumentationTraceability Tests; PaperPipelineTests	withdrawn with justification
A10	spin_chemistry.py; README.md	_lindblad; propagate_enco unter_reference	Figures 5, 7, and 8; Tables 2-3	EncounterDynamicsTests; I ndependentEncounterValida tionTests	superseded by corrected model

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Panel C. Implementation and verification (continued)

Requirement id	Exact file	Exact code object	Figure or table	Test	Evidence status
A11	README.md; sensitivity_analysis.py; paper_analysis.py	encounter_result_row	Figures 2, 4-8; Tables 4-6	PaperPipelineTests	corrected with evidence
A12	spin_chemistry.py; README.md	hamiltonian; EncounterParameters	Tables 1-3	InitialStateAndHamiltonianTests; AuthorityAndModeTests	partially addressed
A13	paper_analysis.py; sensitivity_analysis.py	run_sweep	Table 1	PaperPipelineTests	superseded by corrected model
A14	spin_chemistry.py; tests/ test_paper_pipeline.py	active six-state API	Table 1	PaperPipelineTests.test_active_files_do_not_reference_removed_two_qubit_api	superseded by corrected model
A15	paper_analysis.py	main	Figures 1-11; Tables 1-10	PaperPipelineTests	corrected with evidence
B1	README.md; paper_analysis.py	CAPTIONS; _model_overview_rows	Figure 1; Table 1	PaperPipelineTests	partially addressed
B2	README.md; paper_analysis.py	_model_comparison_rows; _circuit_rows	Figure 1; Tables 1 and 7	PaperPipelineTests	partially addressed
B3	spin_chemistry.py; paper_analysis.py	execute_three_qubit_unitary_circuit; _circuit_rows	Figure 11; Table 7	CoherentValidationTests	partially addressed
B4	configs/parameter_provenance.csv; paper_analysis.py	_provenance_rows	Table 2	AuthorityAndModeTests; PaperPipelineTests	corrected with evidence
B5	paper_analysis.py; plotting.py	_solver_validation; _runtime_memory_benchmark	Figures 8 and 11; Tables 6, 7, and 10	PaperPipelineTests	partially addressed

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B6	paper_analysis.py; plotting.py	main; CAPTIONS	Figures 1-11	PaperPipelineTests.test_everypaper_pipeline_is_not_onblank_and_has_source_csv	corrected with evidence
B7	paper_analysis.py; table_rendering.py	TABLE_PRESENTATION; render_table_from_csv	Tables 1-10	PaperPipelineTests	corrected with evidence

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